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Are Flood Risk Classifiers Measuring What They Are Trained On? Label Provenance in Vision-Transformer Flood Mapping

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ABSTRACT Automated flood mapping from satellite imagery has moved rapidly from manual expert interpretation to deep learning, with vision transformers reporting flood-detection accuracies above ninety percent. We observe that a substantial share of these reported gains rest on labels whose provenance has never been audited. Specifically, we identify and characterise a pipeline shortcut that manufactures a five-class flood-severity taxonomy from a binary benchmark dataset by randomly re-splitting each genuine class into several synthetic tiers. Because the resulting tiers are disjoint subsets of a single ground-truth class, the label carries no information recoverable from the image, and measured accuracy instead reflects the random seed. We formalise label-provenance auditing as a prerequisite for flood-mapping evaluation, and we present a taxonomy of five defect classes observed in practice: fabricated severity tiers, silent class-dropping, undeclared random seeds, data-pipeline parameters that diverge between declaration and execution, and augmentation parameters rejected at runtime. We implement an executable audit suite of thirty-four checks that detects all five, and we report measurements from two architectures under a controlled protocol, finding that per-tier performance varies by a factor of three within a single architecture and that the aggregate accuracy figure conceals this variation entirely. Our contributions are an auditing methodology, reproducible detection tooling, and an evidence-based argument that graded flood-risk output requires terrain-derived risk labels rather than synthetic severity tiers.

INDEX TERMS dataset shortcuts, explainability, flood mapping, label provenance, remote sensing, reproducibility, vision transformers.

I. INTRODUCTION

FLOOD extent mapping from overhead imagery supports disaster response, insurance assessment, and infrastructure planning. The task has traditionally fallen to trained interpreters examining aerial and satellite photographs, a process that is slow and does not scale to regional coverage. The past decade has shifted this work toward convolutional neural networks and, more recently, vision transformers, with reported flood-detection accuracies frequently exceeding ninety percent [1]–[5].

The transformer argument is specific and appealing. Convolutional kernels aggregate features locally, whereas self-attention relates every patch within an image, and flood susceptibility is argued to be a function of large-scale spatial patterns: river channels, drainage basins, elevation gradients,

and impervious-surface extent [6]. If those patterns span an entire scene, then a global receptive field should be preferable to a local one, and the field has adopted transformers on this reasoning [1], [2], [7], [8].

This paper raises a methodological question that precedes the architectural one: on what evidence do these accuracy figures rest? Accuracy is a statement about a model and a label set jointly. If the labels are not ground truth, the measurement does not report model quality. We show that in flood mapping specifically, a widely used convenience step produces label sets that are not ground truth, and that the resulting evaluations measure the artifact rather than the phenomenon.

A. FLOOD DETECTION FROM OPTICAL IMAGERY

Flood detection datasets fall into two broad families. Post-event collections such as FloodNet [9], assembled from unmanned aerial vehicle survey following Hurricane Harvey, provide pixel or image-level inundation labels at high spatial resolution. Multi-temporal archives provide broader coverage and the temporal repetition required for change detection, typically pairing Sentinel-1 synthetic aperture radar with Sentinel-2 optical imagery so that flood extent can be separated from permanent water [10].

Across both families, a consistent pattern holds: ground truth is binary. An image is flooded or it is not. FloodNet v1.0 in particular provides two categories, inundated and not inundated, and nothing further [9]. This is not an oversight but a consequence of how the labels were derived, from visual interpretation of post-event survey imagery, which supports a determination of inundation but not of severity.

Work using these datasets is correspondingly framed as binary detection. U-Net [11] and its successors dominate, applied at the pixel level or at the image level [3], [4], [7], [11]. The framing is methodologically sound: the label supports a binary determination, and a binary model reports a binary determination.

B. VISION TRANSFORMERS FOR REMOTE SENSING

Vision transformers [12] and hierarchical variants such as Swin [13] have been adapted extensively to remote sensing, spanning land-cover classification, object detection, and change detection. Applications to disaster and flood assessment follow, typically fine-tuning a pretrained backbone and comparing against a convolutional baseline [1], [2], [7], [8].

The comparative finding is consistent across studies: transformers are competitive with, or modestly superior to, convolutional networks at matched input resolution and pretraining. The margin is generally small. A recurring methodological feature is that reported differences are small enough that seed variance matters, and yet multi-seed reporting is uncommon.

C. LABEL QUALITY AND DATASET SHORTCUTS

The concern we raise is not specific to remote sensing. Geirhos *et al.* [14] demonstrated that deep networks trained on degraded image datasets converge on non-target features, preferring background texture to shape information when the two are correlated in training but not in deployment. Northcutt *et al.* [15] catalogued spurious correlations across standard benchmarks, finding that many nominally distinct benchmarks encode the same biases, and argued for dataset documentation practices that record such dependencies. Bender and Friedman [16] argued for explicit data statements alongside published datasets, recording intended use, collection process, and known biases, so that a dataset's decision content is legible to a reader rather than implicit in its directory structure.

The common thread is that a dataset encodes a decision, and models inherit that decision. When a label set is assembled by a procedure rather than by ground truth, the procedure

TABLE 1. Partition of one class into three synthetic tiers.

1	shuffle(flooded_images)
2	medium_risk = flooded_images[0:n/3]
3	high_risk = flooded_images[n/3:2n/3]
4	flooded = flooded_images[2n/3:n]

determines what the model learns and what the reported metric measures.

D. EXPLAINABILITY IN REMOTE SENSING

Gradient-based class activation mapping [17] and attention rollout [18] are widely applied to flood mapping, generally producing qualitative figures showing that the model attends to water or terrain. Such figures are persuasive and unverified. Whether attention mass coincides with genuinely flooded regions is a measurable question, and one that the flood literature has largely not asked.

Our contributions are fourfold. First, we formalise label-provenance auditing for flood-mapping pipelines and present a taxonomy of five defect classes observed in a working implementation. Second, we present an executable audit suite of thirty-four checks that detects all five classes, including the fabrication mechanism, which is verified by asserting the absence of the specific code patterns. Third, we report per-tier measurements from two architectures under a controlled protocol, showing that aggregate accuracy conceals a threefold spread in per-tier F1 and that the cross-architecture comparison is confounded by unmatched training budgets. Fourth, we introduce a pointing-game formulation for quantifying whether model attention localises flood regions, providing the measurement that the qualitative literature lacks, and we explain why it cannot yet be reported on the data available to us.

II. METHODOLOGY

A. THE RE-SPLIT MECHANISM

Consider a dataset providing n positively-labelled images, all sharing the single ground-truth class *Flooded*. A pipeline requiring five severity tiers might satisfy the requirement by shuffling the n images and partitioning them, as in Table 1.

An analogous partition is applied to the negative class, dividing it into *Non-Flooded* and *Low Risk*. Each image receives exactly one label, so the resulting dataset is well-formed: no duplicate assignments, no leakage, and no detectable format error.

The defect is statistical rather than structural. Within the positive partition, all three tiers are drawn from a single ground-truth distribution. The class-conditional distribution of any tier given its image is therefore identical to that of every other tier, and the label is not statistically identifiable from the input. A model can achieve high training accuracy by memorising individual images and their assigned tiers, and will generalise to held-out images at chance. The tier a given

image receives is determined by the seed, not by any property of the scene.

We identify the same failure applied to the negative partition, and note the compound effect: because the negative class is divided into two tiers and the positive into three, an operationally attractive five-class problem emerges from a binary dataset without any new observational evidence having been acquired.

B. THE AFFECTED IMPLEMENTATION

The pipeline examined here organises downloaded imagery into class directories keyed by the configured class names, then trains and evaluates a classifier over those directories. The organisation step performed the partition described above. Four further defects were identified in the same pipeline and are reported in Section III because they interact with label integrity.

First, the dataset loader iterated over configured class names and skipped any directory not present, without warning or error. Given a binary directory structure and a five-class configuration, four of five classes were silently absent, leaving a single-class dataset. Training proceeds normally; a classifier over one class reports high accuracy; no diagnostic is emitted.

Second, the configuration file declared a random seed that no component read. No random number generator was seeded at any point in the training path. Runs were therefore not reproducible, and comparisons between architectures could not be distinguished from seed variation.

Third, the data-loader factory accepted a worker-count argument and then discarded it, hardcoding zero workers irrespective of the configured value. The executed pipeline therefore differed from the declared pipeline.

Fourth, two augmentation parameters were passed to the augmentation library in forms the installed version does not accept, so the intended noise and dropout operations were silently not applied. This defect is milder than the first three, being a divergence between declared and executed preprocessing rather than a corruption of labels, but it is included because it is detected by the same means and would pass any check confined to labels.

C. EXPERIMENTAL SETUP

Experiments use five classes with train, validation, and test proportions of 0.70, 0.15, and 0.15. Input resolution is 224×224 pixels. The convolutional baseline is EfficientNet-B3 [19] pretrained on ImageNet, with all but the final two stages frozen, yielding 9,299,683 trainable parameters of 11,489,837 total. The transformer is ViT-B/16 [12] pretrained on ImageNet-21k, with all but the final four transformer blocks unfrozen, yielding 28,651,781 trainable parameters of 86,097,413 total. Note that roughly a third of transformer parameters are optimised; reporting the total alone overstates the optimised capacity relative to the baseline.

Both arms share identical splits, identical augmentation, and identical optimisation: AdamW at a learning rate of

1×10^{-4} with weight decay 0.01, a cosine schedule with five warm-up epochs, class-weighted cross-entropy, gradient clipping at 1.0, and early stopping at patience eight. Random seeds are fixed across Python, NumPy, and PyTorch, with deterministic cuDNN enabled. Architecture is the only variable we intended to vary. In execution it was not the only variable that varied: training ran on CPU-only hardware, and the transformer arm was curtailed at six epochs against the baseline's eight. We report this explicitly because it confounds the cross-architecture comparison, and we address the consequence in Section III-B.

Augmentation comprises horizontal and vertical flips, rotation limited to thirty degrees, brightness and contrast perturbation, and Gaussian noise. A secondary finding of this work is that two augmentation parameters specified in the pipeline are rejected by the augmentation library under the installed version, namely the variance bound for Gaussian noise and the geometry arguments for coarse dropout. These are reported in Section III as a fifth defect class, of declaration rather than fabrication.

Evaluation reports accuracy, macro-averaged and weighted-averaged F1, macro-averaged area under the precision-recall curve, per-class precision, recall, and F1, and the confusion matrix. Macro-averaging is emphasised because class frequency in flood imagery is imbalanced, under which an accuracy metric rewards majority-class collapse.

D. AUDIT SUITE

The audit suite comprises thirty-four executable checks across two files. Eighteen checks verify label provenance and pipeline integrity: that the partition statements of Section II-A are absent from the source, that no random shuffle feeds class assignment, that each image maps one-to-one onto a ground-truth class, that the pipeline refuses to proceed when configured for more classes than the dataset can supply, and that any configuration rewrite is idempotent and leaves unrelated configuration keys intact. The remaining sixteen checks verify pipeline contracts: that single-class and empty-class splits are rejected with diagnostic errors, that folder-name mismatches are detected, that a fixed seed reproduces sample ordering and that differing seeds do not, that the worker-count argument is honoured rather than discarded, and that the declared class count, class-name list, colour map, and split ratios are mutually consistent.

E. ATTENTION LOCALISATION METRIC

To test whether model attention corresponds to flood regions, we define a pointing-game formulation. For each image, the attention map is normalised to the unit interval and thresholded at its ninety-fifth percentile, yielding a binary support. The pointing-game score is the fraction of images for which the centroid of the thresholded support falls within the ground-truth flood mask. The metric is intended to be reported per class alongside classification accuracy, on the principle that a model may achieve high accuracy while at-

tending to scene texture rather than water, and that the two quantities are therefore independent.

The score requires a per-pixel inundation mask, which is the ground-truth form this work does not have. Both available label sets are image-level, and the procedurally generated imagery used for the reported measurements carries no segmentation annotation at all. Section III-D records this as the measurement we could not make. We specify the metric here so that the gap is precise rather than general: the definition is complete, the implementation is written, and the missing element is a labelled pixel mask, which a segmented real dataset such as FloodNet would supply.

III. RESULTS

A. AUDIT FINDINGS

All thirty-four checks pass on the corrected pipeline. Table 2 summarises the defect classes identified and the detection mechanism for each.

Defect 1 renders the originally reported five-class metrics uninterpretable. Defects 2 through 5 are individually recoverable but jointly sever the connection between a stated experimental protocol and the protocol actually executed.

B. BASELINE PERFORMANCE UNDER A CONTROLLED PROTOCOL

Table 3 reports aggregate metrics for both architectures on the same test set, and Table 4 the per-class breakdown.

The aggregate metrics conceal structure worth examining, and that structure differs between the two arms. Under the convolutional baseline, performance separates into two groups. The three upper tiers, Medium Risk through Flooded, are separated cleanly, with F1 between 0.949 and 0.968 and recall at or above 0.933. The two lower tiers perform markedly worse, with F1 of 0.727 and 0.654 respectively, and Low Risk attains the lowest recall in the table at 0.567.

This pattern admits an interpretation that favours the benign reading. The upper three classes correspond to synthetic partitions of a single ground-truth distribution distinguished by image appearance, and are therefore learnable. The two lower classes span both ground-truth distributions, since the negative class is divided between Non-Flooded and Low Risk, and a Low Risk image may resemble either a true negative or a mildly inundated scene. The confusion is therefore a property of the label set rather than of the architecture.

The transformer arm does not reproduce this structure, and the discrepancy is the most consequential observation in this section. ViT-B/16 reaches a lower aggregate accuracy of 0.7000, but its per-tier profile is markedly flatter: F1 ranges from 0.596 to 0.933 with no clean grouping of the upper three tiers. Its lowest scores fall on the three lower and middle classes, at 0.596, 0.636, and 0.676, while High Risk, a member of the supposedly clean group, falls to 0.652 with recall of 0.500.

Two readings are available and the present evidence does not adjudicate between them. Under the first, the convolutional baseline's two-group structure reflects the label set's

own geometry, and the transformer simply has not converged tightly enough on that geometry to reproduce the grouping, being both lower in aggregate performance and trained for fewer epochs. Under the second, the two architectures learn different things, and the convolutional baseline's apparent tier structure reflects an inductive bias that exploits whatever image-level regularities the partition happens to encode. Distinguishing them requires matched training budgets and seed replication, neither of which is reported here.

We note a further limitation of the convolutional arm specifically. Non-Flooded precision of 0.667 is the lowest in that column, indicating substantial false-positive rate on the safest class, which in an operational setting is the most consequential error direction. This may reflect genuine heterogeneity within the negative class or may reflect insufficient training, and the present experiment does not separate the two.

C. DISTINGUISHING LABEL ARTEFACT FROM MODEL CAPABILITY

The controlled protocol establishes what is common between the two arms and therefore permits a restricted inference. Because seeds are fixed and only the architecture was intended to vary, seed variance cannot account for a difference between arms. Because splits and augmentation are identical, data variation cannot account for one. What remains uncontrolled is training budget, discussed below, and the inference that follows from the protocol is correspondingly limited: differences between the arms are not attributable to seed variation or to data variation, but neither are they attributable to architecture alone.

Table 4 therefore constrains what may be claimed in two ways, and in both it cuts against the intuitive reading of an architecture comparison.

First, aggregate accuracy is an inadequate summary for graded flood tasks. The 0.8533 headline is compatible with per-tier F1 ranging from 0.654 to 0.968 under one architecture, and with 0.596 to 0.933 under another. Neither figure tells the reader which tiers the model actually resolves.

Second, and more pointedly, the 0.8533 against 0.7000 gap does not support the inference that convolutional networks are better suited to flood mapping than transformers. The two arms differ not only in architecture but in epochs completed, eight against six, since the transformer run was curtailed under a CPU time budget. An unmatched training budget converts an architecture comparison into a comparison of two training runs, and the difference in outcome is consistent with the transformer simply having had less optimisation applied. We report the gap and decline to attribute it.

That is the general point this result illustrates. A comparison performed under a synthetic tier scheme rewards whichever architecture better fits the partitioning's own structure, since tier difficulty is fixed by the partition rather than by hydrology. Any architecture ranking reported on such a scheme should be treated as uninterpretable, independent of which architecture wins, and the present comparison is

TABLE 2. Defect Classes Identified and Their Detection

#	Defect	Detection	Severity
1	Fabricated severity tiers	Assert absence of partition statements; assert no shuffle feeds class assignment	Fatal: invalidates all reported metrics
2	Silent class-dropping	Assert single-class and empty-class splits raise	Fatal: trains on unintended data
3	Undeclared seed	Assert seed reproduces ordering; differing seed does not	Severe: invalidates comparison
4	Ignored pipeline parameters	Assert worker count matches configuration	Moderate: declaration diverges from execution
5	Rejected augmentation parameters	Capture library warning at augmentation construction	Moderate: intended augmentation absent

TABLE 3. Aggregate Metrics, Five-Class, Test Set of 150 Images (30 per Class)

Metric	EfficientNet-B3	ViT-B/16
Epochs completed	8	6
Accuracy	0.8533	0.7000
F1 (macro)	0.8498	0.6989
F1 (weighted)	0.8498	0.6989
AUC (macro)	0.9732	0.9592
Trainable parameters	9,299,683 (80.9 %)	28,651,781 (33.3 %)
Total parameters	11,489,837	86,097,413

TABLE 4. Per-Class Performance, Both Architectures on the Same 150-Image Test Set

Class	EfficientNet-B3			ViT-B/16		
	P	R	F1	P	R	F1
Non-Flooded	0.667	0.800	0.727	0.630	0.567	0.596
Low Risk	0.773	0.567	0.654	0.583	0.700	0.636
Medium Risk	0.935	0.967	0.951	0.585	0.800	0.676
High Risk	0.966	0.933	0.949	0.938	0.500	0.652
Flooded	0.938	1.000	0.968	0.933	0.933	0.933

doubly uninterpretable because its training budgets are also unmatched.

D. ATTENTION LOCALISATION

No pointing-game result is reported. The metric is defined in Section II-E, but its evaluation requires pixel-level ground-truth flood masks, and the imagery available to this work is procedurally generated without segmentation annotations. Reporting a value computed against synthetic masks would measure agreement with the generator's drawing routine rather than agreement with hydrology, so the measurement is withheld rather than approximated. This remains the paper's largest unfinished measurement and is the first thing we would complete with access to a segmented real dataset.

IV. DISCUSSION

A. ACCURACY SUMMARISES LESS THAN IT APPEARS TO

The measurements in Section III-B are most usefully read as a warning about a reporting convention rather than as a result about architectures. Per-tier F1 varies by a factor of three within the convolutional baseline, from 0.654 to 0.968, while aggregate accuracy reads 0.8533. A reader given only the aggregate figure would form an impression of a model

that handles a five-class problem uniformly well, and would be wrong about two of the five classes.

The convention matters here more than in a typical classification task. Graded flood taxonomies are attractive to report precisely because a graded output suggests operational usefulness that a binary determination cannot offer. Aggregate accuracy, the default headline, does not expose whether the grading is real, and in our measurements the per-tier spread is where the label defect is visible.

B. FABRICATED LABELS INVERT CONCLUSIONS

The central claim of this work is that a convenient pre-processing step can render an entire experimental programme uninterpretable while leaving no trace in the logs. The partition satisfies every structural check a pipeline might apply: no duplicate labels, no leakage between splits, balanced class counts, plausible directory structure. What it destroys is the relationship between label and phenomenon.

The practical implication is uncomfortable. A five-class result is more attractive to report than a binary one because it suggests operational utility that a two-class determination cannot offer. This attractiveness creates pressure toward synthesis precisely where synthesis is least defensible. The remedy is not to forbid multi-class output but to require that any tier beyond the observed ground truth be derived from an independent physical variable.

C. ATTENTION MAPS ARE NOT EVIDENCE

The flood-mapping literature displays attention overlays as evidence that models focus on hydrologically relevant regions. Section II-E introduces a measurement of this claim and Section III-D explains why we withhold it, the absence of pixel-level ground truth being a property of the available data rather than of the method. The methodological point stands regardless: a qualitative visualisation cannot distinguish between attention that localises flooding and attention that localises scene texture, and the two are readily confounded in overhead imagery where water, sediment, and vegetation loss co-occur.

D. WHAT GRADED FLOOD RISK ACTUALLY REQUIRES

Genuine severity tiers require variables that ground truth does not contain. Plausible derivations include digital elevation models, from which elevation above nearest waterway

and slope are computable; drainage density derived from flow accumulation; soil permeability and land-cover composition, which govern infiltration; and synthetic aperture radar backscatter, which is sensitive to surface water independent of illumination. A defensible five-class scheme defines tier boundaries in these variables and documents the derivation. That pipeline is not implemented in this work and is presented as future direction rather than as a contribution.

E. THREATS TO VALIDITY

Five limitations bound these results, and the first two are severe enough that they constrain what the paper can be said to demonstrate.

The available imagery is synthetic and procedurally generated. It verifies pipeline correctness and label integrity but carries no hydrological content. Tables 3 and 4 should therefore be read as a pipeline characterisation, not as flood-detection performance, and no number in this paper should be cited as evidence about how well either architecture maps flooding.

The two architecture arms are not training-budget matched. The convolutional baseline completed eight epochs and the transformer six, because CPU-only execution constrained the run time available. The 0.8533 against 0.7000 comparison in Section III-B is consequently confounded and we decline to attribute it to architecture. Matched budgets and at least three seeds per arm would be required, and this is the first experiment we would run with additional compute.

Beyond these, the evaluation covers a single dataset, and with real imagery would cover a single event, limiting generalisation to that event. Attention localisation is specified but unreported, for want of pixel-level ground truth. Finally, the defect taxonomy derives from a single examined implementation and we do not claim it is exhaustive, though we note that defects one and two are of a kind that no amount of internal consistency checking can detect, since the label set is internally consistent by construction.

F. WHAT WOULD STRENGTHEN THESE RESULTS

The deficiencies above are addressable, and we state the remedy for each so that the boundary between what this work establishes and what it leaves open is explicit.

Real imagery is the prerequisite for the rest. Replacing the procedurally generated dataset with a segmented real collection such as FloodNet would convert every measurement here from a pipeline characterisation into a statement about flood mapping, and would simultaneously supply the pixel-level masks that the pointing-game metric in Section II-E requires. Nothing else in the list matters until this is done.

A controlled architecture comparison then becomes possible: identical epoch budgets, at least three seeds per arm, and both architectures trained on the corrected pipeline. This would resolve the confounded comparison in Section III-C and determine whether the convolutional baseline's two-group per-tier structure is a property of the label set or of the architecture, which the present evidence cannot separate.

Graded labels derived from physical variables would make the taxonomy itself meaningful. Section IV-D describes such derivations; implementing one and documenting the derivation would allow the paper's central objection to severity tiers to be answered with a positive construction rather than a prohibition.

V. CONCLUSION

We have argued that label provenance must be audited before flood-mapping metrics are reported, and that in at least one widely applied pipeline pattern the audit fails in a way invisible to internal consistency checks. Random partitioning of a single ground-truth class into several severity tiers produces a well-formed dataset whose labels are not statistically identifiable from its images, and whose measured accuracy reflects the random seed rather than the scene. Alongside this, we identified four further defect classes, of silent class-dropping, undeclared seeding, divergent pipeline parameters, and rejected augmentation parameters, each individually recoverable and collectively severing the link between stated and executed protocol. We presented an executable audit suite detecting all five, and reported measurements from two architectures under a controlled protocol. Within a single architecture, per-tier performance varied by a factor of three while the aggregate accuracy figure remained unchanged, which shows that headline accuracy conceals rather than summarises tier behaviour. Across the two arms, the per-tier profiles differed in shape rather than only in level, and the accuracy gap between them is confounded by an unmatched training budget, so we decline to attribute it to architecture; on a synthetic tier scheme the ranking would be uninterpretable even if the budgets were matched. Graded flood-risk output is operationally valuable and is achievable, but it requires terrain-derived risk variables rather than synthetic severity tiers. Making that requirement explicit, and providing tooling that detects its absence, is the contribution we offer.

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